

### Assignment 1

**Marking:** Assignment 1 has 4 questions, each worth 10 marks, for a total of 40 marks. Question 4 differs between MATH3303 and MATH7333, so please attempt the version for your course.

**Submission:** Please submit online via blackboard.

**Help:** You are encouraged to discuss assignments in groups, but your solutions must be written in your own words. You are also encouraged to complete assignments **without the use of AI**.

**Question 1.** Let  $G_n = ((\mathbb{Z}/n\mathbb{Z})^\times, \times)$ . In lectures we assumed that  $n$  is prime, but if we take all congruence classes modulo  $n$  that are relatively prime to  $n$ , then  $G_n$  is a group for all positive integers  $n$ . Determine the order of  $G_{p^k}$ , where  $p$  is prime.

**Question 2.** Prove that the smallest non-abelian group has order 6.

(Hint: (1) Show that any group  $G$  with  $|G| \leq 5$  is an abelian group ; and (2) give an example of a non-abelian group of order 6. )

**Question 3.** Let  $H$  and  $K$  be normal subgroups of a finite group  $G$ . Show that if

$$H \cap K = \{1_G\} \quad \text{and} \quad |H||K| = |G|,$$

then  $G \cong H \times K$ .

**Question 4. (MATH3303 only)** Let  $G$  be a group with subgroups  $N$  and  $K$  of index 2, such that  $N \neq K$ .

- (a) Prove that  $[N : N \cap K] = 2$ .
- (b) Prove that  $G/(N \cap K) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ .

(Hint: use the second isomorphism theorem for (a), and the first isomorphism theorem for (b).)

**Question 4. (MATH7333 only)** Let  $G$  be the subgroup of  $\text{GL}_3(\mathbb{R})$  consisting of matrices of the form

$$\begin{bmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix}, \quad \text{where } a, b, c \in \mathbb{R}.$$

- (a) Determine the centre  $C(G)$ .
- (b) Prove that  $G/C(G) \simeq \mathbb{R}^2$ .